EXPERIMENT NO. 7 (Group B)

Aim: Interacting with Web APIs Problem Statement: Analyzing Weather Data from OpenWeatherMap API Dataset: Weather data retrieved from OpenWeatherMap API

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Outputs:

Step - Retrieve Weather Data

In [48]:
```
import requests
import pandas as pd

# Enter your own OpenWeatherMap API key
api_key = "7f5f5d43e25926fc90bc8d6faf684673"

# Coordinates for Pune
lat = 18.184135
lon = 74.610764

# API URL
api_url = (
    f"https://api.openweathermap.org/data/2.5/forecast"
    f"?lat={lat}&lon={lon}&appid={api_key}"
)

# Send request
response = requests.get(api_url)

print("Status Code:", response.status_code)

weather_data = response.json()

# Check whether the request was successful
if response.status_code != 200:
    print("Error:", weather_data)
else:
    print(weather_data.keys())
    print(weather_data["list"][0])
```

```
Status Code: 200
dict_keys(['cod', 'message', 'cnt', 'list', 'city'])
{'dt': 1791309600, 'main': {'temp': 300.04, 'feels_like': 300.52, 'temp_min': 300.0
4, 'temp_max': 300.04, 'pressure': 1014, 'sea_level': 1014, 'grnd_level': 951, 'humi
dity': 51, 'temp_kf': 0, 'dew_point': 289.06}, 'weather': [{'id': 802, 'main': 'Clou
ds', 'description': 'scattered clouds', 'icon': '03n'}], 'clouds': {'all': 41}, 'win
d': {'speed': 1.23, 'deg': 339, 'gust': 2.43}, 'visibility': 10000, 'pop': 0, 'sys':
{'pod': 'n'}, 'dt_txt': '2026-10-06 18:00:00'}
```

Step 2- Extract Weather Data

In [49]:
```
timestamps = [
    pd.to_datetime(item["dt"], unit="s")
    for item in weather_data["list"]
]

temperature = [
    item["main"]["temp"]
    for item in weather_data["list"]
]

humidity = [
    item["main"]["humidity"]
    for item in weather_data["list"]
]

wind_speed = [
    item["wind"]["speed"]
    for item in weather_data["list"]
]

weather_desc = [
    item["weather"][0]["description"]
    for item in weather_data["list"]
]

# Create DataFrame
weather_df = pd.DataFrame({
    "Timestamp": timestamps,
    "Temperature": temperature,
    "humidity": humidity,
    "wind_speed": wind_speed,
    "weather_description": weather_desc
})

# Set Timestamp as index
weather_df.set_index("Timestamp", inplace=True)

# Display first 5 records
weather_df.head()
```

Out[49]:

Timestamp	Temperature	humidity	wind_speed	weather_description
2026-10-06 18:00:00	300.04	51	1.23	scattered clouds
2026-10-06 21:00:00	299.78	50	2.68	scattered clouds
2026-10-07 00:00:00	298.45	48	2.41	few clouds
2026-10-07 03:00:00	300.77	40	3.34	clear sky
2026-10-07 06:00:00	305.87	28	4.65	clear sky

Step 3 — Clean Data and Convert Temperature

In [50]:
```
# Handle missing values
weather_df.fillna(0, inplace=True)

# Convert Kelvin to Celsius
weather_df["Temperature"] = weather_df["Temperature"] - 273.15

# Find maximum and minimum temperature
print("Max Temp (°C):", weather_df["Temperature"].max())
print("Min Temp (°C):", weather_df["Temperature"].min())

# Display the data
weather_df
```

```
Max Temp (°C): 35.34000000000003
Min Temp (°C): 25.100000000000023
```

Out[50]:

Timestamp	Temperature	humidity	wind_speed	weather_description
2026-10-06 18:00:00	26.89	51	1.23	scattered clouds
2026-10-06 21:00:00	26.63	50	2.68	scattered clouds
2026-10-07 00:00:00	25.30	48	2.41	few clouds
2026-10-07 03:00:00	27.62	40	3.34	clear sky
2026-10-07 06:00:00	32.72	28	4.65	clear sky
2026-10-07 09:00:00	34.27	23	4.26	clear sky
2026-10-07 12:00:00	34.50	23	3.82	clear sky
2026-10-07 15:00:00	30.80	28	5.79	overcast clouds
2026-10-07 18:00:00	28.35	38	4.54	overcast clouds
2026-10-07 21:00:00	27.52	42	1.16	overcast clouds
2026-10-08 00:00:00	25.10	50	2.09	scattered clouds
2026-10-08 03:00:00	27.56	48	3.38	clear sky
2026-10-08 06:00:00	32.77	33	3.87	clear sky
2026-10-08 09:00:00	34.95	25	4.20	clear sky
2026-10-08 12:00:00	34.25	25	2.38	few clouds
2026-10-08 15:00:00	32.17	28	3.14	broken clouds
2026-10-08 18:00:00	30.35	35	0.67	broken clouds
2026-10-08 21:00:00	28.64	45	0.72	broken clouds
2026-10-09 00:00:00	26.65	53	0.36	broken clouds
2026-10-09 03:00:00	29.02	43	3.85	few clouds
2026-10-09 06:00:00	32.93	29	3.96	clear sky
2026-10-09 09:00:00	34.77	23	2.09	few clouds
2026-10-09 12:00:00	34.59	22	2.20	scattered clouds
2026-10-09 15:00:00	29.27	52	6.68	broken clouds
2026-10-09 18:00:00	28.37	54	4.44	broken clouds
2026-10-09 21:00:00	27.57	54	3.90	broken clouds
2026-10-10 00:00:00	25.46	64	3.18	broken clouds
2026-10-10 03:00:00	28.01	48	2.15	clear sky
2026-10-10 06:00:00	33.22	29	4.28	clear sky

Timestamp	Temperature	humidity	wind_speed	weather_description
2026-10-10 09:00:00	35.14	22	3.18	few clouds
2026-10-10 12:00:00	34.46	22	3.09	few clouds
2026-10-10 15:00:00	32.12	27	3.00	overcast clouds
2026-10-10 18:00:00	29.55	42	1.76	overcast clouds
2026-10-10 21:00:00	27.81	45	2.21	broken clouds
2026-10-11 00:00:00	25.63	45	3.58	scattered clouds
2026-10-11 03:00:00	28.45	40	3.41	clear sky
2026-10-11 06:00:00	33.16	29	4.88	clear sky
2026-10-11 09:00:00	35.34	22	4.78	scattered clouds
2026-10-11 12:00:00	33.59	22	7.47	broken clouds
2026-10-11 15:00:00	29.67	30	6.00	scattered clouds

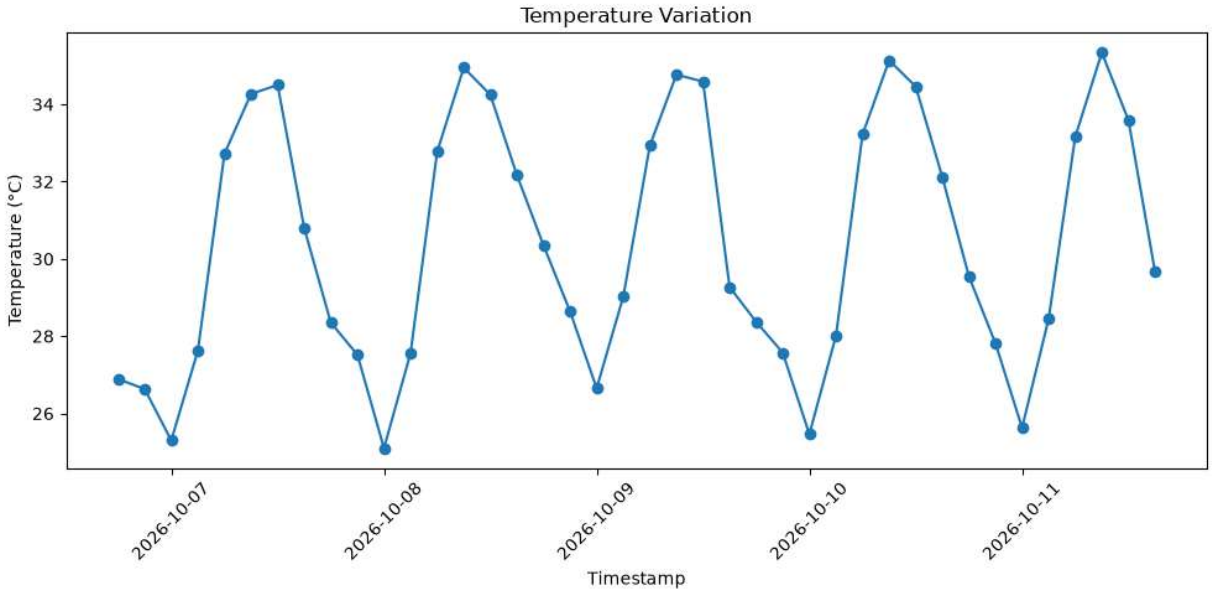
In [51]:
```
import matplotlib.pyplot as plt

plt.figure(figsize=(10, 5))

plt.plot(
    weather_df.index,
    weather_df["Temperature"],
    marker="o"
)

plt.xlabel("Timestamp")
plt.ylabel("Temperature (°C)")
plt.title("Temperature Variation")
plt.xticks(rotation=45)
plt.tight_layout()

plt.show()
```



In [54]:
```
import folium

weather_map = folium.Map(
    location=[50.5, 1.5],
    zoom_start=5
)

# London marker
folium.Marker(
    [51.5074, -0.1278],
    popup=folium.Popup(
        """
        <b>London</b><br>
        Temperature: 21.7°C<br>
        Humidity: 46%
        """,
        max_width=200
    ),
    tooltip="London",
    icon=folium.Icon(color="red", icon="cloud")
).add_to(weather_map)

# Paris marker
folium.Marker(
    [48.8566, 2.3522],
    popup=folium.Popup(
        """
        <b>Paris:</b><br>
        21.7°C,<br>
        Humidity<br>
        46%
        """,
        max_width=150
    ),
    tooltip="Paris",
    icon=folium.Icon(color="red", icon="cloud")
).add_to(weather_map)
```

```
# Save
weather_map.save("weather_map.html")

print("Weather map created successfully!")
print("Name:Aditya Rade Roll No:BEAD23167")
```

```
Weather map created successfully!
Name:Aditya Rade Roll No:BEAD23167
```